

misinformation is less likely (Altay et al. (2020)), and when such reputational concerns are missing, even calling out individuals sharing misinformation is fairly ineffective (Mosleh et al. (2021a)).

Finally, this proposition provides a possible pathway for low-reliability content (often containing misinformation) to become viral. Vosoughi et al. (2018) argued that misinformation spreads farther, faster, deeper and more broadly than truthful news on social media. This evidence was criticized by Grinberg et al. (2019) who showed that, once the effects of sensational news items is controlled for, misinformation does not spread faster than truthful content. Propositions 1 and 2 provide a rationalization of these patterns. All else equal, misinformation does not spread faster than truthful content as in Grinberg et al. (2019). However, because, as observed in Molina et al. (2021) and Kozyreva et al. (2020), sensational content is often low-reliability, these two propositions together imply that misinformation may be more likely to become viral. Whether this happens or not depends on the boost from sensationalism. When this is limited, low-reliability articles containing misinformation spread less because of concerns of users that others will call them out for sharing this content. But when this sensationalism boost is high, misinformation can become viral. Additionally, the strategic complementarity in sharing decisions implies that sufficiently sensational misinformation can become viral very rapidly, because once an individual thinks others are going to share this sensational item, she becomes much more likely to share herself, even if she has doubts about its veracity.

4 Island Networks and the Implications of Homophily

In this section, we will conduct comparative statics with respect to the sharing network $\mathbf{P}$. Throughout this section, we take the sharing network as given, and then return to how it is shaped by the algorithms of social media platforms in Section 5.

The focus on comparative statics with respect to the sharing network necessitates two modifications from our analysis so far. *First*, we will restrict attention to *island networks* (or equivalently, the stochastic block model), which are lower-dimensional than general networks we have allowed so far. Namely, in an island network, agents are partitioned into k blocks of size $N_1, N_2, \dots, N_k$, called *islands* each with some constant (but not necessarily equal) share of the population N . Each agent i has a type $\ell_i \in \{1, \dots, k\}$ corresponding to which block (or “island”) she is in. Link probabilities are then given as:

$$p_{ij} = \begin{cases} p_s, & \text{if } \ell_i = \ell_j \\ p_d, & \text{if } \ell_i \neq \ell_j \end{cases}$$

where $p_s \geq p_d$. Without loss, we assume each of the islands is weakly connected.

Furthermore, we assume the prior distribution for agents on the same island ℓ is the same, and is denoted by H_ℓ . We also assume that islands are ranked according to their belief distributions. In particular, each island ℓ has distribution H_ℓ with support on $[b^{(\ell)}, b^{(\ell+1)}]$, where $1 \geq b^{(1)} > b^{(2)} > \dots > b^{(k)} > b^{(k+1)} \geq 0$.¹³ This implies that lower-indexed islands have stronger right-wing beliefs.

¹³This assumption is adopted for simplicity. Our results generalize if we instead assume that these distributions are ranked in terms of first-order stochastic dominance: $H_1 \succeq_{FOSD} H_2 \succeq_{FOSD} \dots \succeq_{FOSD} H_k$. However, this generalization requires considerably more formalism and notation, motivating our focus on disjoint supports.